

KHALIL BEN ALI

— Full-Stack & AI Engineer

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SUMMARY

Full-stack engineer (5+ years) shipping TypeScript platforms, .NET desktop apps and production AI. Primary engineer of an AI operations hub integrating 10 SaaS systems for a US financial firm; sole developer of an AI brand experience for JDE's L'OR. Owns systems end to end — schema to WebGL, prompt to print queue.

EXPERIENCE

nBold — Full-Stack & AI Engineer · 2024 → present · remote (Paris HQ)

nBold builds Microsoft Teams workspace-templating & governance SaaS (ISO 27001, SOC 2).

- Designed and built the platform's Teams meeting-intelligence feature end to end (~11,000 lines, 67 files, 321 tests): Microsoft Graph transcript capture with a net-new app-only auth path, AI summarization, and CRM-aware LLM routing to Teams Adaptive Cards — durable workflows proven crash-safe.
- Primary engineer of Cuddy Brain, a Teams-embedded ops hub mirroring 10 SaaS systems into PostgreSQL (~115 tables) — ~3,860 commits, ~69% of backend history (NestJS, BullMQ/Redis, React, Azure OpenAI).
- Shipped AI call intelligence: call → transcription → Azure OpenAI summary → extracted follow-up tasks → Teams Adaptive Card, minutes after hang-up, with single-flight locking and crash recovery.
- Built AI monitoring (obituary adjudication with deterministic post-gates; address-change detection over public records and client comms) and co-built a LangGraph agent compiling client reviews from 7 systems.
- Hardened the platform: shared Azure OpenAI client (timeouts, retries, usage tracking), HMAC webhook verification, secret-scanning CI, caching and N+1 performance overhauls.
- Sole developer of L'OR UNLIMITED REMIX (JDE × HLWN 2026), an AI brand app + event kiosk: Gemini with deterministic fallback, on-device CLIP vision, 60fps WebGL fluid engine, offline-first PWA, Bluetooth thermal printing — 129/129 commits, ~24.6k LOC, 11 weeks to production.
- Built Cuddy Scanner, a .NET 9/WPF TWAIN document scanner (barcode batch separation, PDF assembly, SharePoint filing) in daily production use.

Freelance — Software Developer · 2023 → 2024

- Children's game app (React Native, Expo, Redux) and cross-platform travel planner (RN + MobX; Express/MongoDB backend; React admin).

SolvR — Software Developer · 2023

- Refactored and shipped a cross-platform chat app (React Native, Expo, MobX, Firebase): real-time messaging, secure auth, state architecture.

Tynass IT — Software Developer · 2019 → 2022

- Delivered AR tourism app, job-seeker platform, fashion e-commerce and corporate sites (React, Material-UI, Express, MongoDB).

SKILLS

TypeScript · React/Next.js · React Native · NestJS/Node · PostgreSQL · Redis/BullMQ · Azure OpenAI · Gemini · LangGraph · RAG (pgvector) · transformers.js · C#/.NET 9 · WPF · C++20/JUCE · Microsoft Graph & Teams · Salesforce · Docker · GitHub Actions · Vercel

SELECTED SIDE PROJECTS

- HyperMesh DSP & HyperGpu (C++20/JUCE): LAN mesh for real-time DSP offload and a GPU-accelerated plugin host; CI builds for 5 platforms.
- Assurances BIAT KB: explainable RAG chatbot (NestJS, pgvector, Ollama) with cited answers and RBAC audit trail.

EDUCATION

B.A. Multimedia Communication, Higher Institute of Multimedia Arts of Manouba (ISAMM), 2019

LANGUAGES

Arabic (native) · English (professional) · French (A2-B1) · Italian (A2)